MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

WEEK 6 TEAM PROGRESS REPORT

Memory, State and Interoperability

PROJECT: University Student-Support Case Agent

Reporting period: 5–9 October 2026

Compiled: 13 October 2026, after the Week 6 integration merge

Repository: https://github.com/Kabandaarthur/BSE4104.git

TEAM MEMBERS

Name	Role
Kabanda Arthur	Project / Requirements Lead
Tumukunde Kato Andrew	Application / Integration Lead
Ariko Sossy Joel	AI Engineering Lead
Garanga John	Quality / Security & DevOps Lead

1. Deliverables

Deliverable	Status	Evidence
State model, memory design and data-handling note	Done	`docs/architecture/Week6_State_Model_and_Contract.pdf` defines the one approved field, lifecycle, expiry, reset, isolation and approval boundary.
State contract and session recovery tests	Done	`tests/test_state_contract.py`, `tests/test_session_store.py`; persistence, expiry, corruption recovery, isolation and approval-gate behavior.
Session manager and persistent local store	Done	`src/session_store.py`; JSON persistence, atomic replacement, corruption quarantine and 24-hour expiry.
Session-aware `/chat` and reset integration	Done	`src/main.py`, `src/orchestrator.py`; cookie-backed active-case context, reset endpoint, and session-aware turn wrapper.
MCP capability specification and adapter prototype	Done	`docs/architecture/MCP_CAPABILITY_SPEC.md`, `src/mcp/`; stdio JSON-RPC interface for the two Week 4 tools.
MCP compatibility and interoperability report	Done	`tests/test_mcp_adapter.py`, `docs/evaluation/MCP_INTEROPERABILITY_REPORT.md`; schema parity, input validation, error mapping and client roundtrip.
Working memory/state demonstration	Done	Automated `/chat` integration test creates an examination ticket and re-reads its actual status in the same cookie session.
Week 6 progress report	Done	This report.

2. Contributions

Member	Contribution	Evidence
Kabanda Arthur	Defined the approved state model, session lifecycle, authorization boundary and traceability; wrote the executable state contract.	`Week6_State_Model_and_Contract.pdf`; `tests/test_state_contract.py`; commit `72dd26b`.
Tumukunde Kato Andrew	Implemented the file-backed SessionStore, expiry/reset/recovery behavior, cookie sessions and `/chat` integration.	`src/session_store.py`; `src/main.py`; `tests/test_session_store.py`; PR #22.

Member	Contribution	Evidence
Ariko Sossy Joel	Defined the MCP interface and built the adapter, strict input validation, schema mapping, error mapping and compatibility tests.	`MCP_CAPABILITY_SPEC.md`; `src/mcp/`; `tests/test_mcp_adapter.py`; `MCP_INTEROPERABILITY_REPORT.md`; PR #23.
Garanga John	Integrated session context with the bounded orchestrator and added multi-turn session regression tests.	`src/orchestrator.py`; `tests/test_session_chat.py`; PR #24.

3. Demonstration and Interoperability

The state flow is scoped to a server-issued cookie session. A successful ticket or case tool result supplies the remembered identifier; the follow-up still calls `get_case_status` for the current status. The deterministic approval-gate test creates an `EXAMINATION` ticket, confirms it is `PENDING_STAFF_APPROVAL`, and verifies the later same-session lookup remains pending. Session data does not contain approval state or message history.

The MCP adapter exposes `get_case_status` and `create_support_ticket` using the internal tool schemas and shared dispatch/error behavior. Its stdio client test covers initialization, tool listing, a successful call and rejection of invalid arguments. The adapter remains a prototype for the mock store; it does not provide authentication or per-student authorization at the MCP boundary.

No live-model Week 6 `/chat` traces were found under `evidence/traces/`. The working demonstration and approval check are deterministic automated tests, not live Gemini traces.

4. Verification

Check	Result	Evidence
Week 6 state, session and MCP test slices	**66 passed**	`pytest -q -rs tests/test_state_contract.py tests/test_session_chat.py tests/test_session_store.py tests/test_mcp_adapter.py`
Full repository test suite	**379 passed, 5 skipped**	`pytest -q -rs`, run 13 October 2026
Live-model tests	**5 skipped**	Require `RUN_LIVE_TOOL_TESTS=1` and `GEMINI_API_KEY`.
Warnings	**1 deprecation warning**	Starlette test client references the deprecated `anyio.abc.BlockingPortal` alias.

The full suite initially exposed a Windows-only failure in the MCP subprocess test: Windows cannot use `select()` on a subprocess pipe. The test now reads stdout on a background thread and passes on this Windows environment. Sessions remain opt-in, and the approval-gate contract test now exercises the real `/chat` session path and ticket handlers with the model/orchestrator decision stubbed.

5. Open Issues and Week 7 Readiness

1. **Audit logging is still outstanding.** The state-model PDF requires reads, writes and clears of `active_case_id` to be loggable with a timestamp and session ID. `src/session_store.py` currently logs storage errors and an ignored write, but not each read/write/reset. Add structured, privacy-conscious audit events before using real student records.
2. **Capture live application traces.** The Week 6 plan assigns live-model trace coverage, but no Week 6 traces are checked in and five live-model tests remain skipped without credentials. Capture and retain representative same-session follow-up and session-isolation traces when the approved model credentials are available.
3. **Correct a traceability path in the state-model PDF.** Its MCP implementation row names `src/mcp_adapter.py`; the implementation is under `src/mcp/`.
4. **Keep the MCP authentication limit explicit.** The current prototype targets the mock store. Do not connect it to real student data until identity and per-student authorization are enforced.